Quoc Thang Nguyen

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Education

San Jose State University — San Jose, CA
Bachelor of Science in Software Engineering — *Expected May 2027*

Mission College — Santa Clara, CA
A.A.S. Computer Information Systems, GPA: 3.96; Liberal Arts: Natural Science & Mathematics — *May 2024*

Experience

Advanced Packaging Technology Development – Software Intern — May 2026 – Aug 2026
Micron Technology — *Boise, ID*

- Built and shipped a reusable **computer-vision image-quality system** in **Python**, **OpenCV**, and **NumPy** for **semiconductor inspection**, progressing from an experimental prototype to a feature integrated into Micron's internal semiconductor image-processing workflow.
- Developed automated focus and image-quality filtering using **multi-metric sharpness analysis**, segmentation, and calibrated thresholds, achieving **92%+ accuracy** on large labeled inspection datasets and generalizing to previously unseen images.
- Iterated through **27 pipeline versions** and validated results using **PCA**, clustering, robust z-scores, geometric descriptors, and cross-dataset testing; generated browsable galleries and engineering reports to accelerate review.

Full Stack Developer — Oct 2023 – Feb 2024
Powers Engineering and Inspection, Inc. — *San Jose, CA*

- Built an inspection management platform using **SvelteKit**, **Node.js**, **PostgreSQL**, and **Prisma**, supporting onboarding, scheduling, and asset-tracking workflows.
- Migrated **50K+ records** from **MySQL** to **PostgreSQL** using **Flyway** and enforced schema constraints to improve data integrity.
- Implemented **Role-Based Access Control (RBAC)** with hierarchical role delegation for multi-level user permissions.

Machine Learning Engineer Intern — Jun 2023 – Jul 2023
Powers Engineering and Inspection, Inc. — *San Jose, CA*

- Trained a **CNN** using **TensorFlow/Keras** to classify industrial liquid-storage tank images, achieving **70% model accuracy**.
- Developed a **Flask** backend with Dropbox API integration and automated image preprocessing and synchronization using **Redis workers** and cron jobs.
- Containerized the application with **Docker** and deployed it to **AWS Elastic Beanstalk** through **Amazon ECR**.

Backend Developer - Intern — Jun 2022 – Oct 2022
Covet Health — *San Jose, CA*

- Developed **RESTful APIs** using **Spring Boot**, **Spring Data JPA**, and **Hibernate** for a healthcare platform.
- Built PostgreSQL-backed CRUD workflows for patients, physicians, insurance, and medications and used **Docker** for isolated testing.
- Automated CI/CD using **Bitbucket Pipelines**, deploying services to **AWS Elastic Beanstalk**.

Projects

Particle Collision System | *Verlet Integration, Sweep-and-Prune* | Live Demo

- Built a **2D physics simulation** using Verlet Integration with broad-phase Sweep-and-Prune collision detection.

ClaimIt – Campus Lost & Found | *Next.js, Firestore, Firebase, GCS* | Live Demo

- Built a **real-time** campus lost-and-found platform with authentication, claim verification, admin moderation, live updates, and Google Cloud Storage.

3D Portfolio Website | *React, Three.js* | Live Demo

- Developed an interactive 3D portfolio using **React**, **Three.js**, custom shaders, and optimized 3D models.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C, SQL
ML / Computer Vision: OpenCV, NumPy, scikit-learn, TensorFlow, Keras, CNNs, PCA, Clustering, Image Processing, Feature Engineering
Frameworks / Databases: React, Next.js, SvelteKit, Node.js, Flask, Spring Boot, Prisma, PostgreSQL, MySQL, Firebase
Tools / Cloud: Docker, Git, AWS, Redis, Linux, Flyway, Bitbucket Pipelines, CI/CD, Google Cloud Storage